## Q1. What is a Decision Tree? Explain its structure with a real-life example.

A **Decision Tree** is a supervised learning algorithm used for both classification and regression. It works by breaking down a dataset into smaller subsets while simultaneously developing an associated decision tree.

- **Structure**:
    - **Root Node**: The very top node representing the entire dataset, which gets split into two or more homogeneous sets.
    - **Decision (Internal) Nodes**: Nodes where further splits occur based on specific features.
    - **Leaf (Terminal) Nodes**: The final nodes that represent the predicted class or value; they do not split further.
- **Real-Life Example**: Deciding whether to play tennis based on the weather. The **Root** might be "Outlook" (Sunny, Overcast, Rainy). If "Sunny," the **Decision Node** checks "Humidity." If humidity is "Normal," the **Leaf Node** is "Yes" (Play).

## Q2. Gini Impurity vs. Entropy

- **Gini Impurity**: Measures the frequency at which a randomly chosen element from the set would be incorrectly labeled. It is computationally faster because it doesn't involve logarithmic functions.
- **Entropy**: Measures the degree of "disorder" or uncertainty. It uses a log scale, making it slightly more computationally intensive.
- **Scikit-learn Default**: It uses **Gini Impurity** by default because the results are usually similar to Entropy, but Gini is faster to calculate, leading to better performance on large datasets.

## Q3. Overfitting in Decision Trees

**Overfitting** occurs when a tree becomes too complex, capturing noise and outliers in the training data rather than the general pattern.

- **Detection**: You can detect it by comparing accuracies. If **Training Accuracy is very high** (e.g., 99%) but **Testing Accuracy is significantly lower** (e.g., 70%), the model is overfitted.

## Q4. Pruning in Decision Trees

**Pruning** is the process of removing branches that provide little power to classify instances, thereby reducing complexity and overfitting.

- **Pre-pruning**: Stopping the tree growth early (e.g., setting `max_depth` or `min_samples_split`).
- **Post-pruning**: Allowing the tree to grow fully and then removing nodes that do not significantly improve performance.

## Q5. Feature Importance

**Feature Importance** assigns a score to input features based on how useful they are at predicting the target variable.

- **Business Impact**: It helps businesses identify key drivers of success. For example, a bank might find that "Duration of Call" is the top feature, signaling that staff should focus on engaging customers longer to increase conversion rates.